Real course photos

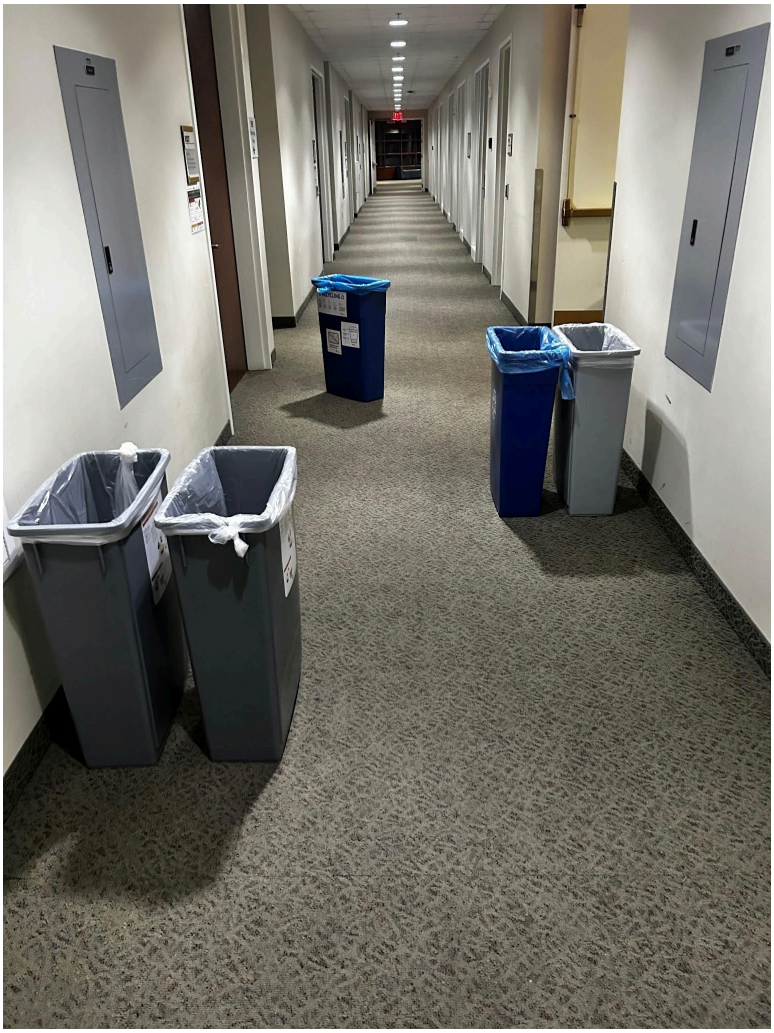


Figure 1: View from course start

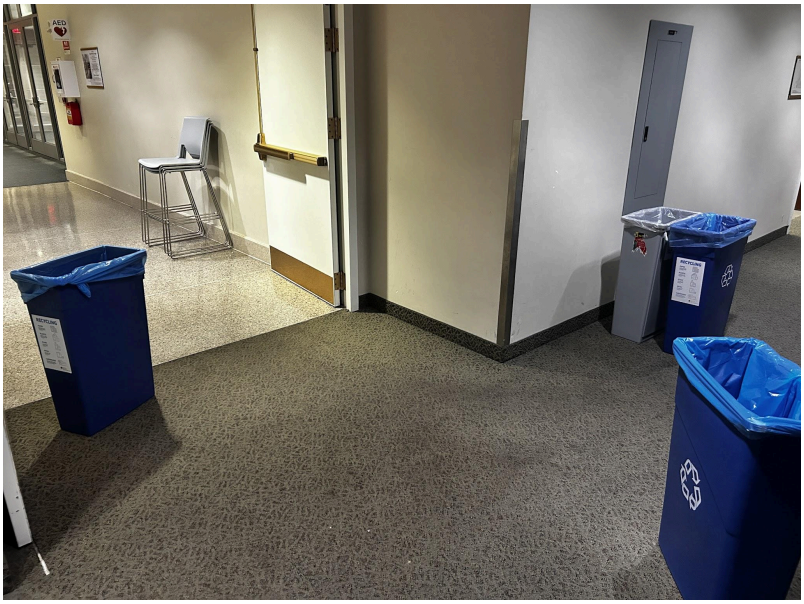


Figure 2: View from middle of the course



Figure 3: View from course finish

Real course details

Same general layout as previous real courses, but with an additional bin in the second hallway section. The hallway and bin measurements from real_report_1 apply.

Final bin -> right side of second hallway: 54”
Second pair of bins -> third bin, horizontal: 52”
Third bin bottom left edge -> left wall, not including the door inset: 16.5”
Third bin top right edge -> corner between first and second hallways (diagonal): 54”
Second bin pair top right edge -> third bin bottom left edge (diagonal): 52”
Second bin pair left edge -> left wall: 54”
First bin pair top -> second bin pair bottom: 40”
First bin pair top right edge -> second bin pair bottom left edge (diagonal): 51”
First bin pair right edge -> right wall: 55”

There were some chairs past the fourth bin in the second hallway area - those were not supposed to be there, but we left them. They’re pretty easy to avoid.

Bag command

ros2 bag record -o real_bag10 /diffbot_base_controller/cmd_vel /cmd_vel_raw /diffbot_base_controller/odom /odometry/filtered/local /odometry/filtered/global /global_costmap/costmap /local_costmap/costmap /velodyne_points_nav /imu/data /plan /odom_goal_pose /odom_goal_pose_rviz /tf /dynamic_joint_states /tf_static /local_costmap/published_footprint /global_costmap/published_footprint /rosout /behavior_tree_log /speed_limit /speed_governor_info

Run #1 (Teleoperation)

A run from start -> finish.
Proves the course is comfortably driveable manually.

Run #2 (Autonomous)

Absolutely fabulous. I noticed a bit of noise from the motors around the third - fourth bin, indicating PID oscillations around the setpoint.

Run #3 (Autonomous)

Driving the course from finish -> start.

Did very well, but a hiccup at the end - it got *very* close to the left wall before stopping and backing up. I may have not given it enough time to recover.

Run #4 (Autonomous)

Another finish -> start run.

Failed between the second and first bin pairs, and ran its full recovery sequence. This illustrates why I don't want to clear the costmap - if the obstacles are underneath the lidar frustum, they go out of view. The robot didn't contact any obstacles when it continued, but that was pure luck.

Run #5 (Autonomous)

Another finish -> start run.

Success this time, but again got very close to the left wall on the final turn past the first bin pair.

Run #6 (Autonomous)

Another finish -> start run.

Very similar to run #5 despite the different starting position.

Run #7 (Autonomous)

A finish -> start run that started from a different entry point to further emphasize that one turn.

Got quite close to the first obstacle when it entered the course.

Run #8 (Autonomous)

A start -> finish run starting from inside our lab. Did very well overall, nothing stood out.

Run #9 (Autonomous)

Finish -> start run from the usual position. Again, got very close to that wall at the end.

Run #10 (Autonomous)

Start -> finish. There was a hiccup at the start; it tried pathing into our lab. It was switched to teleop mode, driven back, and given the goal again with the tester blocking off the alternate route.

Report requests

Same content as the other iteration reports, and answer the following:

- Did the IMU latency changes work?
- Why was finish -> start harder than start -> finish? Why in particular that turn?

In the next iteration, we'll make the course contain more of those kinds of turns.